Classification of waste materials in very-high-resolution satellite imagery

State of the art and thesis proposal

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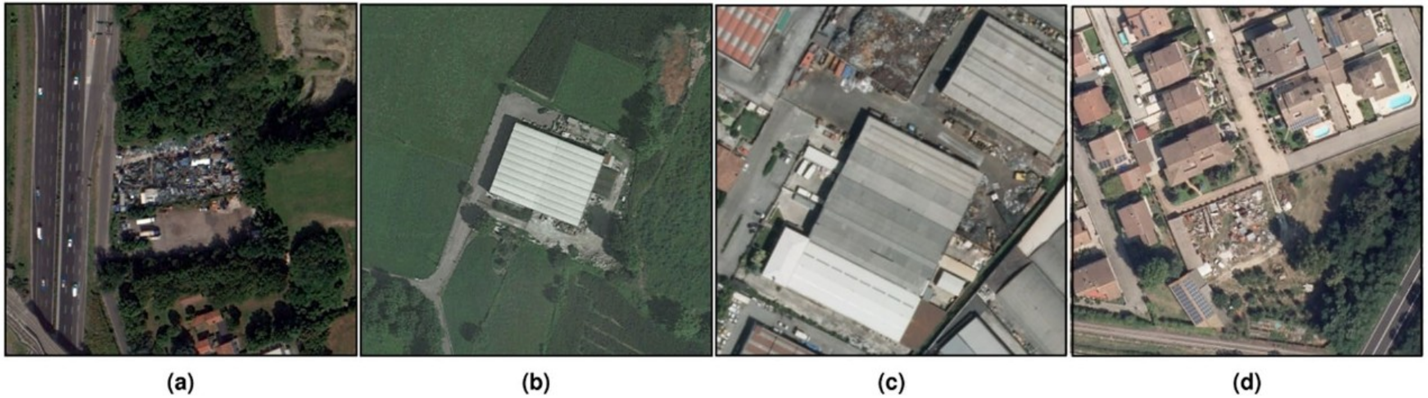
Context

Illegal waste dumping is an environmental crime with direct public-health consequences. Agencies (ARPA) have limited inspection capacity, and priority depends on what is dumped: rubble, plastics and asbestos-cement imply very different hazards.

Detecting dump sites from images is mature. Recognising the material is not: the reference survey (50 works, 1987-2023, almost all RGB) marks it as an open problem.

One thesis in the group has addressed material classification directly (Alari 2024). This work starts from there.



Images: AerialWaste positive examples, Torres and Fraternali 2023 (CC BY 4.0) · survey: Fraternali et al. 2024

Task definition

Objective: given a VHR satellite image of a suspected dump site, output the set of waste materials present.

Task: multi-label classification at image level.

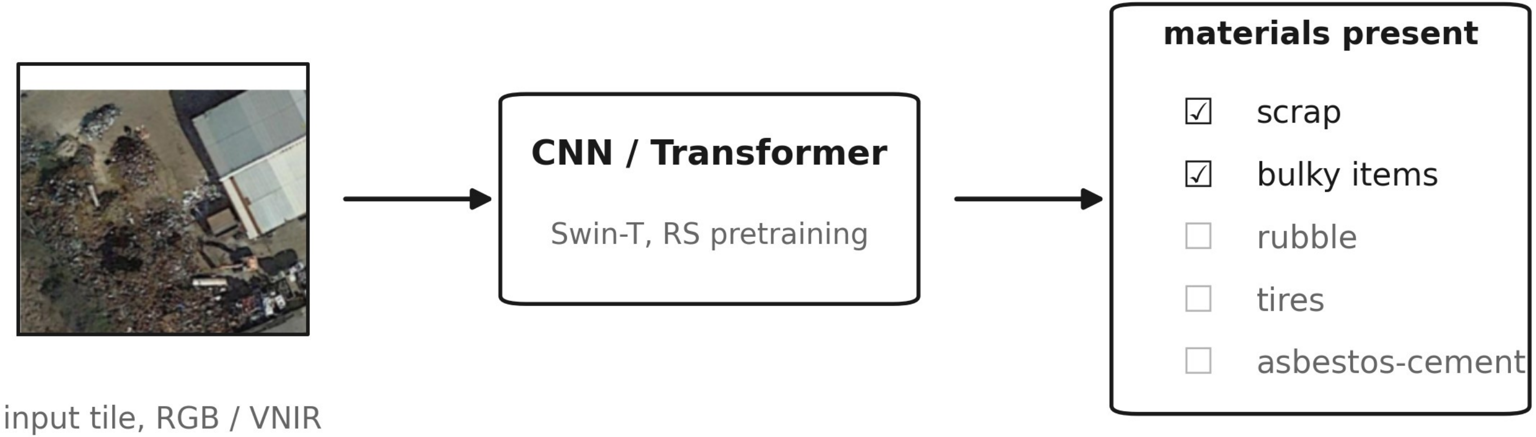
Input: VHR optical imagery, GSD 0.2 to 1.3 m. Aerial RGB for the baseline (AerialWaste); satellite VNIR for the multispectral arm, up to 8 bands. SWIR excluded: not in the planned acquisitions, and its 3.7 m GSD does not match the task. Sentinel-2 excluded: too coarse (10-20 m).

Technique: classification, not object detection or segmentation. Annotations are image-level (Alari 2024: 11,477 multi-label); waste piles have no stable shape for boxes; no segmentation masks exist for these datasets.

Research question: does VNIR input improve waste-material classification over RGB, and for which materials?

Annotations: Alari 2024 · imagery: WorldView-3, Pléiades Neo (VNIR + pan)

Task scheme



One image in, a set of material labels out. The same scheme runs with RGB or VNIR input; only the first layer of the network changes.

Tile: Alari 2024. The multi-label formulation follows the group predecessor.

The material taxonomy

Ten-category grouping used in the group predecessor; the full annotation set counts 13 categories (adds foundry waste, sludge, tanks as separate classes).

Samples and taxonomy: Alari 2024 (politesi 10589/230633), built on AerialWaste annotations

Materials: which are considered and why

Material	Target	Spectral analysis	Reason
Rubble, plastic, wood, tires	yes	yes	frequent and confusable in RGB
Asbestos-cement	yes	yes, dedicated pilot	public regional ground truth (WFS)
Vehicles, tanks, containers, scrap, bulky, big bags	yes	no	shape-based; RGB sufficient in literature
Sludge, foundry waste	yes	no	few labels; visually ambiguous at this GSD

All 13 categories remain classification targets, for continuity with the group dataset. The RGB-versus-VNIR analysis concentrates on the subset where colour is ambiguous and labels are sufficient.

Decision criteria: visual ambiguity in RGB, label availability, hazard relevance

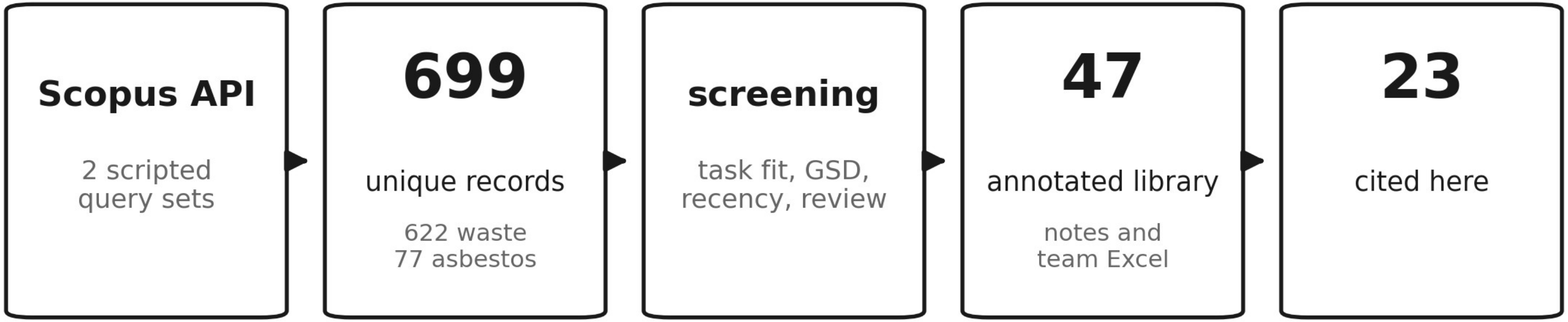
Per-material coverage in the literature

Material	Dedicated works at task scale	Notes
Asbestos-cement	Saba 2026, Bonifazi 2026, Abbasi 2024, Cilia 2015	roofs only, never inside a waste taxonomy
Tanks	Ramachandran 2024, YOLOv7-OT 2024	mature object detection
Vehicles	vehicle-detection DA 2020	VHR object detection
Containers	truck-and-container 2025	size and status classification
Scrap	ELV Hybrid-YOLOv5 2025	close-range infrared, not satellite
Rubble / inert	CWLD 2024; debris volume UAV 2022	C&D segmentation; UAV photogrammetry
Plastic, wood, tires, bulky, big bags	none found in scope	only as classes in AerialWaste / Alari
Sludge, foundry waste	none found	only as classes in Alari

Scope: terrestrial, GSD compatible with the task. This table answers how thin the material-level literature actually is.

Literature search: method and numbers

Scripted queries on the Scopus API (waste detection in remote sensing; asbestos roof mapping), deduplication, manual screening with explicit criteria, snowballing from the Fraternali 2024 survey and the Alari 2024 references.



Search artifacts: papers/literature_search (queries, raw and deduplicated records) · library: papers/notes

What was kept, what was excluded

Excluded group	Examples	Reason
Sentinel-2 and marine debris	MARIDA 2022, Tisza 2023	10-20 m: pixels too mixed; different domain
Spaceborne hyperspectral	Shepherd 2025 (EnMAP), EMIT 2025	material evidence, but 30-60 m GSD
SWIR-based VHR	Aguilar 2021, Aguilar 2025, Zhou 2021	SWIR not in our acquisitions; 3.7 m GSD
Laboratory / conveyor	Vitek 2025, SpectralWaste 2024	not Earth observation
EO foundation models	DOFA 2024, AnySat 2025, Prithvi 2024, +6	pretrained at 10-30 m; sub-metre transfer unproven

Kept: works on terrestrial waste or roof materials at task-compatible GSD, plus the reference spectral library. The excluded papers stay in the annotated library as context.

Full accounting in papers/INDEX.md (47 papers, each with a structured note)

Related work: site-level waste detection

Work (year)	Input / GSD	Task	Method	Result
Gibellini 2025	aerial RGB, 20 cm	classification	Swin-T, RSP pretraining	F1 92.0; cross-region -5.1
AW ensemble 2025 *	aerial RGB	classification	CNN + transformer ensemble	binary F1 92.4
Disaitek 2024	Pléiades Neo, 30 cm	detection	operational service	~95% on sites >2 m2 (vendor)
CascadeDumpNet 2024	Pléiades, 50 cm	object detection	two-stage CNN + AutoML	mAP 84.6, cross-city transfer
Sun 2023	VHR RGB, 0.3-1 m	object detection	BCA-Net (Faster R-CNN)	~2,500 dumpsites, 28 cities
CWLD 2024	GF-2 + GE, 0.5-0.8 m	segmentation	DeepLabV3+ variant	F1 88.9, construction waste
AerialWaste, Torres 2023	aerial RGB, 20-50 cm	dataset + cls.	ResNet-FPN baseline	F1 80.7; 22 material tags

All RGB: they answer where a site is, not what it contains. * preprint; Disaitek is a vendor figure.

AerialWaste: the reference dataset

10,434 locations from ARPA Lombardia records, 487 municipalities; three sources: AGEA orthophotos (20 cm), WorldView-3 (30 cm), Google Earth (50 cm).

Binary site labels for detection, plus 22 material tags (type of visible object + storage mode) annotated by experts.

Material tags cover about 72% of positives, but the dataset was published for detection: material tags are metadata, not a benchmark.

This is the label base both Gibellini 2025 and Alari 2024 build on.

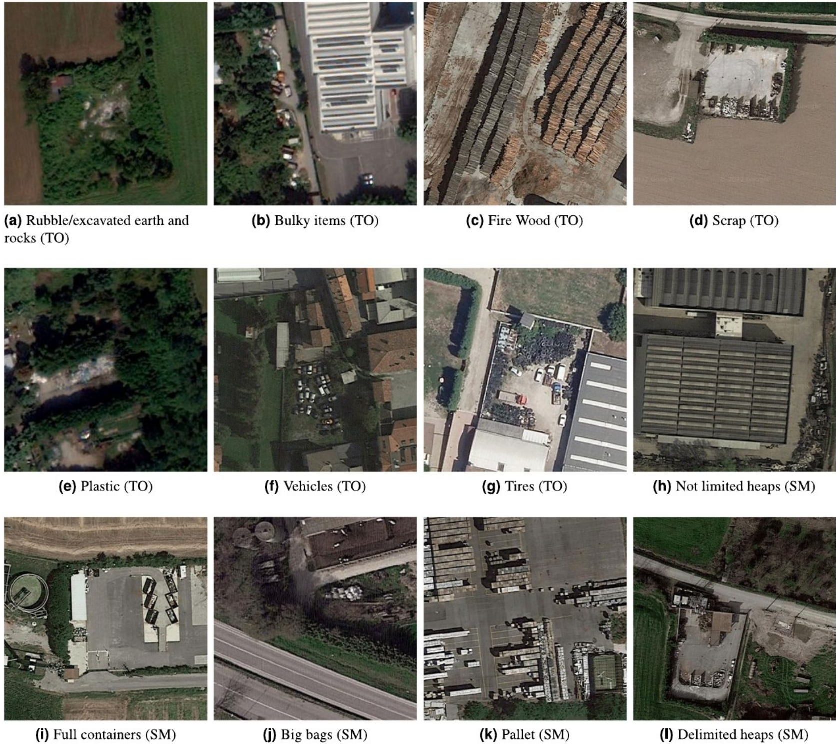


Image: annotated-label examples, Torres and Fraternali 2023, Scientific Data (CC BY 4.0)

Gibellini 2025: the site-level baseline

Binary waste / no-waste classification on AerialWaste; Swin-T with remote-sensing pretraining, two-step fine-tuning.

F1 92.02 in domain. Cross-region generalisation drops 5.1 points on average (Greece 85.4, Serbia 83.8, Romania 91.5).

Saliency maps confirm the model looks at waste heaps, but the output is presence only: no material information.

This architecture and training recipe is the starting point of the proposal.

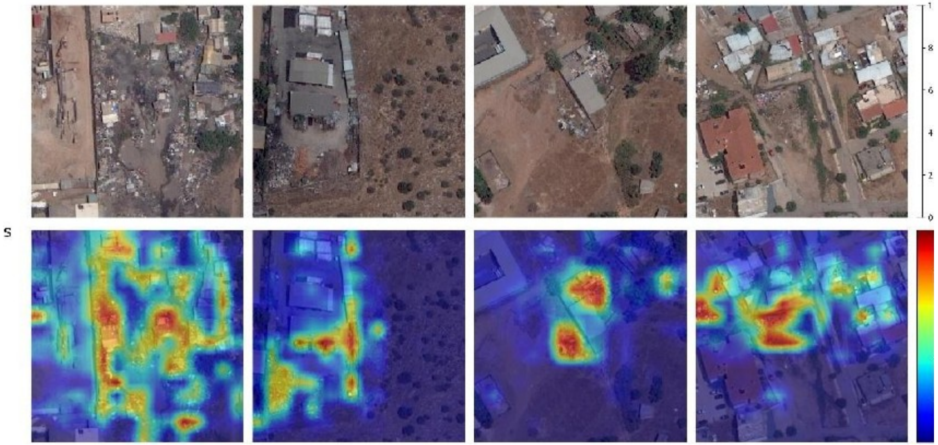


Image: true positives with saliency overlays, Gibellini et al. 2025 (Greek generalisation set)

Alari 2024: the direct predecessor

First work in the group to frame waste-material recognition as multi-label classification.

Dataset: 11,477 multi-label annotations over 13 categories; 3,190 positive and 7,190 negative images, built on AerialWaste imagery and ARPA records.

Models: ResNet-50 and Swin backbones with FPN; three classification-head designs; weighted binary cross-entropy for label imbalance; different pretraining sources compared.

Results: weighted F1 69.21 on five categories, 59.42 on ten. Growing the taxonomy from 5 to 10 costs 9.8 points.

Input is RGB only. The spectral dimension is untouched: that is the opening this thesis takes.

 Alari 2024, M.Sc. thesis, PoliMi (advisor Fraternali, co-advisor Gibellini), politesi 10589/230633

Related work: material-level classification

Work (year)	Input / GSD	Task	Result
Alari 2024 (PoliMi)	satellite RGB	multi-label cls.	wF1 69.2 (5 cat.), 59.4 (10 cat.)
Saba 2026 *	WV-3 VNIR, 1.24 m	pixel cls.	asbestos, Macro-F1 97.6
Bonifazi 2026	WV-3 VNIR+SWIR	pixel cls.	asbestos roofs, removal tracking
Abbasi 2024 *	aerial RGB	OBIA cls.	asbestos, OA ~96 (shape + time)
Cilia 2015	airborne HSI, 3 m	pixel cls.	asbestos, PA 89 / UA 86

One multi-material predecessor; everything else addresses a single material, asbestos, on roofs. No work measures RGB versus VNIR on waste materials.

* paywalled or preprint, reported as such

The asbestos line, in detail

Saba 2026: WorldView-3, VNIR only, 32 classifiers compared; best Macro-F1 97.6 per-pixel. Red Edge and NIR carry the discrimination.

Bonifazi 2026: WorldView-3 multi-temporal workflow; building-level decisions from pixel classification; tracks roof removals between acquisitions.

Abbasi 2024: aerial RGB with temporal features, OA ~96: shape and time can substitute spectra at fine GSD.

Cilia 2015: airborne hyperspectral, PA 89 / UA 86, plus a weathering index from red/NIR: the degradation angle.

Asbestos slate on drone RGB (2023): partially recognisable by shape and texture alone.

Why it matters here: public ground truth (Lombardy WFS, 10,903 roofs) and VNIR evidence make asbestos the natural pilot.

Image: building-level asbestos classification on WorldView-3, Bonifazi et al. 2026, Geomatics (CC BY 4.0)

Related work: objects and close range

Work (year)	Input	Task	Result
Ramachandran 2024	VHR satellite	object detection	tanks, P 0.96 / R 0.97, >169k mapped
YOLOv7-OT 2024 *	VHR satellite	object detection	tanks, precision 95.9
Truck-and-container 2025 *	VHR satellite	object detection	container size and status
Vehicle DA 2020 *	VHR 30-50 cm	object detection	+10% with domain adaptation
ELV Hybrid-YOLOv5 2025	close-range IR	object detection	scrap metals, mAP 84.2
UAV solid waste 2024 *	UAV RGB	segmentation	OA >94, generic waste piles
C&D debris UAV 2022 *	UAV + photogrammetry	segmentation	IoU 0.90 + volume estimation
fCLIPSeg 2025	aerial RGB	segmentation	debris, Dice 0.70, event-agnostic

Shape-defined objects are mature; material composition still needs spectra or close range. * pending full-text verification

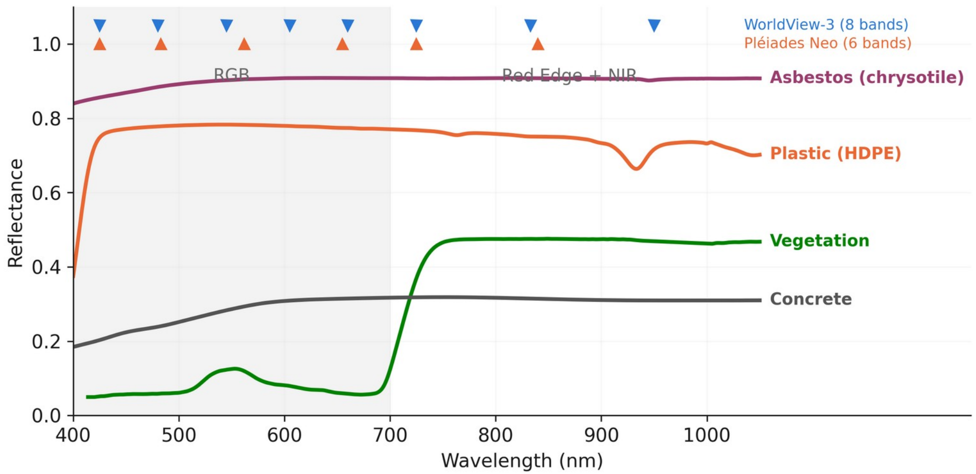
Where RGB falls short, and what VNIR adds

Different materials share the same colour at 0.3-1.3 m: plastic sheets, asbestos-cement and concrete all appear grey.

In the group predecessor, moving from 5 to 10 material categories costs 9.8 F1 points (69.2 to 59.4). Finer material distinctions are the hard part.

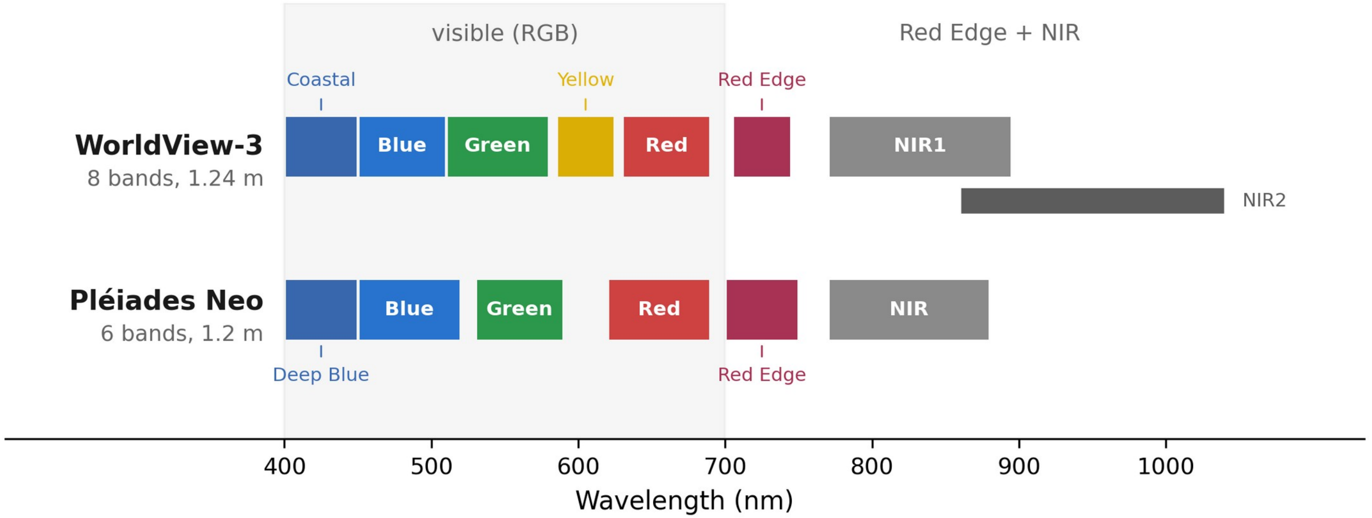
Red Edge and NIR separate vegetation, bare soil and weathered surfaces; in Saba 2026 they drive asbestos discrimination.

Whether this helps waste materials, and which ones, has not been measured. It is the object of this thesis.



Reflectance: USGS splib07a (Kokaly et al. 2017), 400-1050 nm. Band centres: Maxar, Airbus.

Available imagery



Panchromatic: 0.31 m (WorldView-3), 0.30 m (Pléiades Neo). Access: commercial, or free quota via ESA proposal.

This is the band budget of the study: no SWIR, no Sentinel-2. RGB baseline runs on AerialWaste (20-50 cm).

Band ranges: Maxar WorldView-3, Airbus Pléiades Neo data sheets

The in-scope state of the art at a glance

Work	Input	Task	Material info
Fraternali 2024 (survey)	-	survey, 50 works	declares the material gap
AerialWaste 2023	aerial RGB	dataset	22 tags, metadata only
Gibellini 2025	aerial RGB	classification	none (presence)
Alari 2024	satellite RGB	multi-label cls.	13 categories, wF1 59-69
AW ensemble 2025 *	aerial RGB	classification	none (binary)
Sun 2023 / CascadeDumpNet 2024	VHR RGB	object detection	none (sites)
Disaitek 2024 (vendor)	Pléiades Neo	detection	coarse type qualification
CWLD 2024	GF-2 + GE	segmentation	one material (C&D)
Saba / Bonifazi / Abbasi / Cilia	WV-3, aerial, HSI	pixel cls.	one material (asbestos)
Tanks / vehicles / scrap works	VHR, close range	object detection	shape, not composition

Ten lines summarise every in-scope line of work. Material-level evidence: one multi-label predecessor + one material studied in isolation.

What is missing in the literature

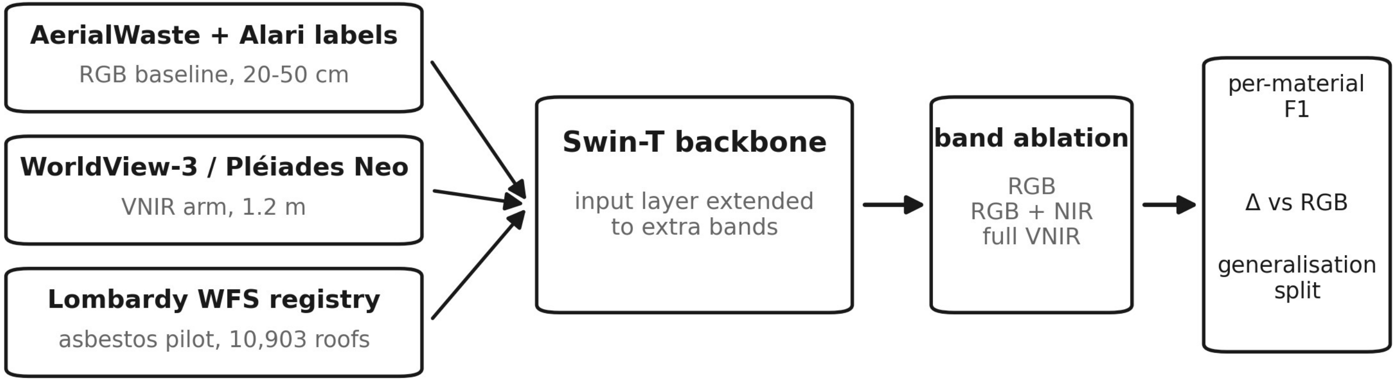
1. Multi-material classification has one direct precedent, with ample margin: wF1 59-69 (Alari 2024).
2. No work measures the added value of VNIR bands over RGB for waste materials at very high resolution.
3. Results are reported as aggregate scores; per-material behaviour is not analysed.
4. Generalisation across regions is rarely evaluated. At site level it costs 5 F1 points (Gibellini 2025).
5. Asbestos is studied on roofs, in isolation, and never inside a waste-material taxonomy.

Each gap maps to one element of the proposal on the next slides.

Proposed work: approach

Multi-label image classification, continuing the group line (Gibellini 2025, Alari 2024). Same taxonomy, input extended from RGB to VNIR.

Band ablation with everything else fixed: same architecture, same splits, only the input changes.



Backbone: Swin-T with remote-sensing pretraining (group baseline); extra bands enter via the input layer.

First step: the asbestos pilot

1. Extract roof polygons from the Lombardy WFS registry (10,903 mapped asbestos-cement roofs) and pair them with the available imagery.
2. Build matched RGB and VNIR inputs for the same roofs; add negative roofs from regional building footprints.
3. Train the same classifier on both inputs; compare F1, precision, recall on held-out areas.
4. Decision point: the measured VNIR delta on a single, well-labelled material tells whether the full multi-label extension is worth the acquisition cost.

Why asbestos first: public pixel-accurate labels, one material, VNIR evidence in literature (Saba 2026), and direct regulatory value.

Ground truth: Regione Lombardia WFS, Mappatura 2020 (EPSG:32632)

Proposed work: evaluation

Per-material F1 alongside weighted and macro averages: aggregates hide exactly the classes this thesis targets.

Delta versus the RGB baseline for each band configuration, with confidence intervals over repeated runs.

Generalisation: train and test on disjoint geographic areas, besides the standard split.

Reference points: wF1 69.2 / 59.4 (Alari 2024); Macro-F1 97.6 (Saba 2026, per-pixel) as upper reference for the pilot, not directly comparable.

If VNIR does not help a material, that is a documented negative result with practical value for sensor choice.

	standard split	**cross-region split**
RGB	baseline F1	baseline F1
RGB + NIR	**ΔF1 per material**	**ΔF1 per material**
full VNIR	**ΔF1 per material**	**ΔF1 per material**

Metrics: per-class F1, macro-F1, weighted F1, confusion matrices. Splits frozen before testing.

References (1/2)

Alari 2024. Fighting environmental crime with deep learning: classifying waste materials from illegal landfills in satellite imagery. M.Sc. thesis, PoliMi, 10589/230633.

Fraternali et al. 2024. Solid waste detection, monitoring and mapping in remote sensing images: a survey. Waste Management / arXiv:2402.09066.

Gibellini et al. 2025. A deep learning pipeline for solid waste detection in remote sensing images. Waste Management Bulletin.

Torres, Fraternali 2023. AerialWaste: a dataset for illegal landfill discovery in aerial images. Scientific Data 10:63.

Sharmily et al. 2025. AerialWaste ensembles (preprint). · Zhang, Ma 2024. CascadeDumpNet. Remote Sensing of Environment 313.

Sun et al. 2023. Revealing influencing factors on global waste distribution. Nature Communications 14:1444.

CWLD 2024. Construction waste landfill dataset. Scientific Data. · Disaitek 2024, vendor report (Airbus).

Saba et al. 2026. J. Hazardous Materials. · Bonifazi et al. 2026. Geomatics 6(3):41. · Abbasi et al. 2024. RSASE. · Cilia et al. 2015. ISPRS IJGI 4(2).

Full annotated library: 47 papers, papers/INDEX.md

References (2/2)

Asbestos slate from drone imagery, 2023 (training-data study).

Ramachandran et al. 2024. Deep learning to map well pads and storage tanks. Nature Communications.

YOLOv7-OT 2024. Storage tank detection for large-scale remote sensing images. Remote Sensing.

Truck-and-container detection from satellite imagery, 2025 (preprint).

Vehicle detection with domain adaptation in VHR imagery, 2020.

ELV Hybrid-YOLOv5 2025. Non-ferrous metal detection in end-of-life vehicles (close-range infrared).

UAV solid-waste segmentation, 2024. · C&D debris volume from UAV photogrammetry, 2022. Drones.

fCLIPSeg 2025. Event-agnostic debris segmentation. · Kokaly et al. 2017. USGS Spectral Library v7 (splib07a). USGS DS 1035.

23 works cited; the remaining library papers are kept as screened context (excluded groups on slide 9)